PROJECT SUMMARY · SOCIAL STATISTICS

Survey Design, a Guttman Scale and a Composite Well-Being Index

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8
survey sections, ~10 minutes

0.912
reproducibility of the final Guttman scale

r = −0.22
aggression vs optimism

0.989
Kendall's W across 5 index methods

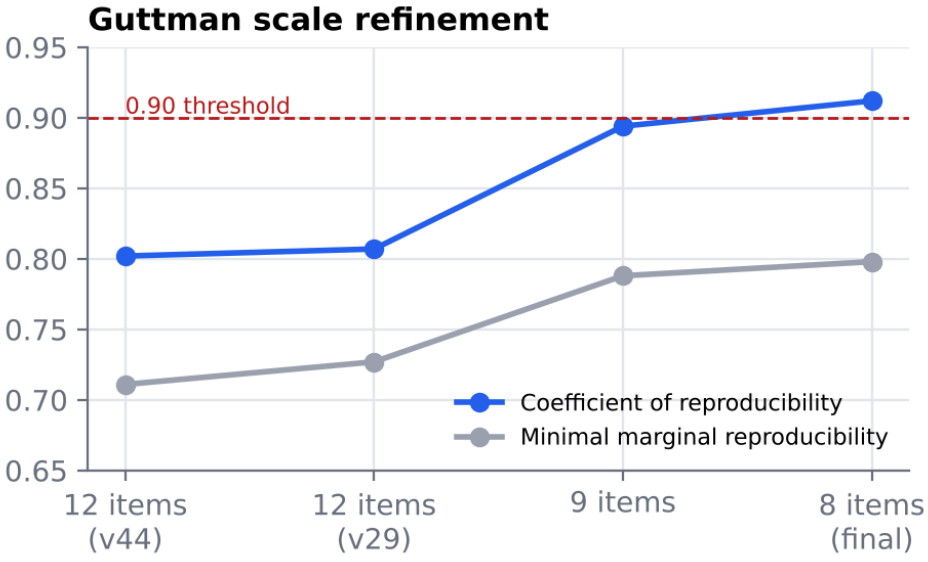
Three applications of quantitative methods to social research: designing a questionnaire, building and validating a measurement scale, and constructing a composite indicator for Italian regions.

1 · Questionnaire design

A survey for Milano-Bicocca students on university catering, intended to inform the renewal of the service contract. The research goal was broken down into an area tree with eight sections (profile, use of the service, dietary needs, food offer, organisation, quality, value for money, overall assessment). Filter questions route users and non-users of the service; four-point ordinal scales without a neutral point are alternated with other formats to limit response sets; neutral wording reduces bias. The questionnaire takes about ten minutes and is anonymous. It was not fielded, so it has not been tested on real respondents.

2 · Guttman scale of reactive aggression

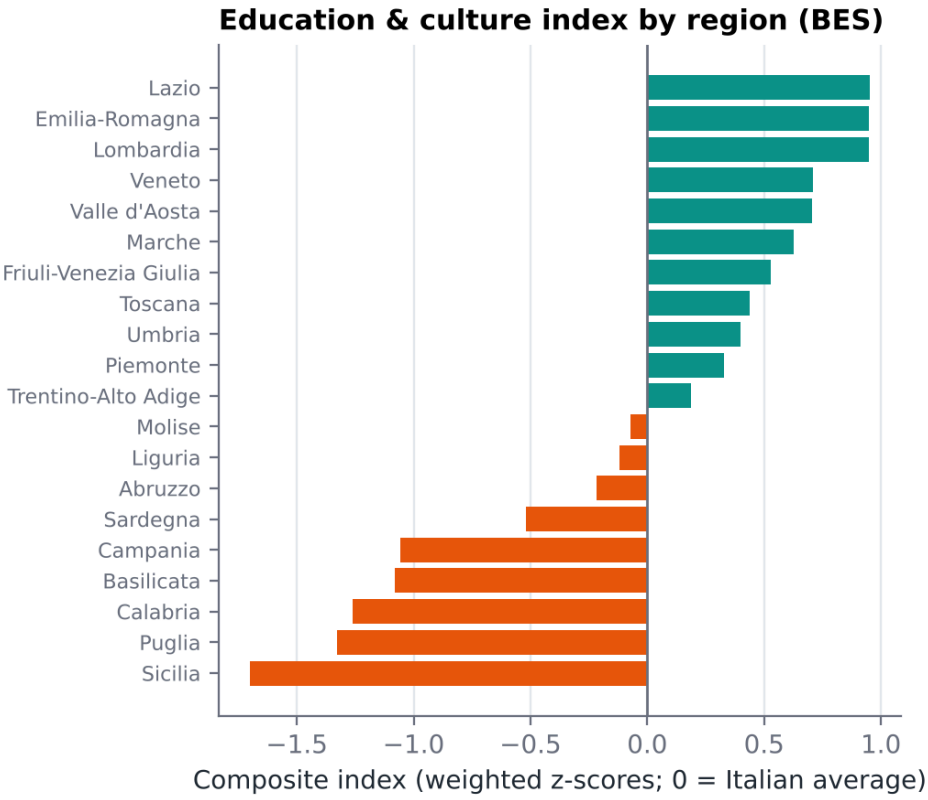
From a 60-item behavioural questionnaire, 12 items on reactions to conflict and frustration were ordered from mild hostility to overt aggression. Items producing the most scaling errors were dropped step by step until the 8-item scale reached a coefficient of reproducibility of 0.912, above the conventional 0.90 threshold. Women score slightly but significantly higher (Welch t-test), mostly on subtler, indirect items. Optimism, measured with the LOT-R, is slightly higher among men (by 1.4 points, $p < 0.001$), and aggression is negatively associated with optimism ($r = -0.22$, 95% CI −0.28 to −0.15, $p < 0.001$): a moderate association, not evidence of causation.



Reproducibility at each stage of item selection.

3 · Composite index of education and culture (ISTAT BES)

Seven BES indicators were grouped into five sub-dimensions (early childhood education, basic skills, cultural participation, digital skills, tertiary attainment), standardised as z-scores and combined with weights of 17–25%. To check that the result did not depend on the method, five aggregation approaches (ranks, index numbers, min–max, z-scores, shares) were compared: all pairwise Spearman correlations exceed 0.98 and Kendall's W is 0.989. Lazio, Emilia-Romagna and Lombardy lead the ranking; Sicily, Puglia and Calabria close it. K-means and hierarchical clustering (elbow method) identify three groups of regions, broadly high, intermediate and low performers, with a clear North–South gradient. The index correlates positively but weakly, and not significantly, with life satisfaction at regional level: objective and subjective well-being capture related but different things.



Composite education and culture index by region.

Limitations

- The questionnaire was not piloted; scales without a neutral point may push respondents towards one side.
- The Guttman scale is only just above the 0.90 threshold and relies on self-reported data, which are prone to social desirability bias.
- Weights involve judgement, and some missing BES indicators required imputation or exclusion.

> English summary of the full report, which is written in Italian (elaborato.pdf). Charts in this summary were redrawn with English labels from the figures reported in the analysis.